



Business Economics & Scaled Cost Projections

A comprehensive breakdown of unit economics, engineering optimizations, and profitability models for scaled market deployment on the Google Play Store.

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LAUNCH TARGET
Google Play Store (15% Commission Tier)

DATA STATUS
Verified (Code Audit)

Executive Summary

Zivofit is a next-generation, AI-driven nutrition and packaging analysis app. Powered by highly optimized, client-directed mobile scanning routines and high-efficiency Cloud Functions, Zivofit enables users to log workouts, track meals via text, voice, or photos, and receive detailed health metrics. By combining local deterministic barcode lookups with optimized multimodal generative AI, the application delivers a premium, data-dense user experience at an operating cost that is 95%+ lower than standard industry implementations.

Core Pricing & Revenue Retention

Zivofit is monetized via monthly and yearly auto-renewing subscriptions on the Google Play Store. Under the 15% commission tier (applicable to developers earning under \$1M annually), Zivofit retains 85% of gross revenue, yielding strong net monthly payments from subscribers.

SUBSCRIPTION PLAN	GROSS RETAIL PRICE	GOOGLE COMMISSION (15%)	NET REVENUE / MONTH	ANNUALIZED NET / USER
Monthly Plan	₹249.00 / month	₹37.35	₹211.65	₹2,539.80
Yearly Plan	₹1,499.00 / year	₹224.85	₹106.18 (eff. monthly)	₹1,274.15

Strategic Pricing Philosophy

The Yearly Plan (priced at ₹1,499, or effectively ₹125/month) is our primary customer acquisition driver, offering the user a 50% discount compared to the Monthly Plan. This locks in 12-month retention upfront and provides high immediate working capital to fund growth, while net retained revenue (₹106.18/user/month) remains orders of magnitude higher than backend operating costs.

Unit Economics Per User Profile

Operating costs are variable and depend on individual user activity. By analyzing real-world behavior patterns, we categorize our subscribers into five profiles based on the daily volume of AI scans (text logs, photo logs, and Zivo Lens analyses). Even under severe usage anomalies, every user segment generates a strong positive contribution margin.

PROFILE	% OF USERS	DAILY ACTIVITY DESCRIPTION	MONTHLY COST	MONTHLY MARGIN (MONTHLY PLAN)	MONTHLY MARGIN (YEARLY PLAN)
Light User	50%	1-2 text logs + 1 barcode (free) + 0-1 Vision scans	₹0.72	99.7%	99.3%
Regular User	35%	2 text + 1 photo log + 1 barcode + 1 Vision scan	₹1.98	99.1%	98.1%
Power User	12%	3 text + 2 photo logs + 2 barcodes + 5 Vision scans	₹6.48	96.9%	93.9%
Extreme User	3%	5 text + 5 photo logs + 2 barcodes + 30 Vision scans	₹33.50	84.2%	68.4%
Abuser (Max Cap)	<1%	Hits daily FUP cap of 75 AI scans every single day	₹47.19	77.7%	55.6%

Variable Cost Drivers

The variable cost consists of two elements: ****Gemini AI API token generation**** (95% of expenses) and ****Firebase cloud infrastructure**** (5% of expenses). Firebase Hosting, Firebase Authentication, and barcode queries to OpenFoodFacts/OpenBeautyFacts are completely free. Infrastructure costs (Firestore database transactions and Cloud Run serverless compute) are zero for 85% of users because their volume fits within Firebase's generous daily free tiers.

The "No-Loss" FUP Cap Safety Net

To prevent automated API attacks or script exploitation, we enforce a daily Fair Usage Policy (FUP) soft cap of 75 total AI scans. As demonstrated above, even if a user maxes out this limit every single day of the month (which requires scanning an item every 13 minutes for 16 hours straight), the absolute maximum cost is capped at ₹47.19/month. Thus, **Zivofit maintains a profitable margin on every active customer with zero negative-margin exposure.**

AI & Infrastructure Cost Optimizations

By implementing client-side pre-processing and server-side constraints, we reduced the baseline cost of an AI scan from ****₹2.00–₹3.00**** down to a current average of ****5–10 paise**** (95%+ cost reduction). This architectural efficiency directly creates our high-margin profile.

ENGINEERING STRATEGY	TECHNICAL IMPLEMENTATION	BEFORE OPT.	AFTER OPT.	COST/RESOURCE IMPACT
Image Pre-Processing	Downscale images to max 1024px and compress to JPEG at 80% quality on-device before backend upload.	3MB – 8MB raw photo files	80KB – 200KB optimized JPEGs	95% savings in cloud storage and upload bandwidth.
Tile Capping (Vision)	Capping image dimensions prevents Gemini from splitting photos into multiple 512x512 tiles.	2k – 4k input tokens/image	258 – 512 input tokens/image	90% reduction in multi-modal input token cost.
Model Selection	Migrated Cloud Functions from heavy models to the lightweight, low-latency Gemini 2.5 Flash-Lite .	₹2.00 – ₹3.00 flat per scan	\$0.10/M input, \$0.40/M output tokens	95%+ reduction in direct cost-per-inference.
Deterministic Bypasses	Client-side local barcode extraction (ML Kit) maps to OpenFoodFacts APIs. Bypasses AI vision entirely.	2,530 tokens / ₹0.022 per scan	0 tokens / ₹0.00	100% savings for products containing recognized barcodes.
Firestore Caching	Completed AI analyses are indexed by barcode. Re-scans read from Firestore rather than invoking Gemini.	6,450 tokens / ₹0.078 per request	0 AI tokens / ₹0.000025 database read	100% AI savings on duplicate scans of the same product.
Selective Decoding	JSON schemas and prompts instruct Gemini to decode <i>only</i> additives of concern, omitting safe ingredients.	800 – 1,500 output tokens	300 – 800 output tokens	50%+ reduction in expensive output generation tokens.

Competitive Moat: Architectural Cost Efficiency

Many consumer AI startups fail because their unit costs grow linearly with engagement. Zivofit's local scanning, caching layer, and image downscaling break this relationship. As our product database grows, Firestore cache hits increase, meaning **average cost per user actually decreases as the user base scales**.

Financial Projections Per 1,000 Active Users

Below is a comparative profit and loss model representing 1,000 active subscribers under two distributions: a **Realistic User Mix** (50% Light, 35% Regular, 12% Power, 3% Extreme) and a **Worst-Case Scenario** (100% of users are Abusers maxing out limits every day on the cheaper Yearly Plan).

Realistic Mix Monthly Profit

97.3% Margin

₹1,03,343

Based on 1,000 users on the Yearly Plan. Total monthly COGS is only ₹2,836.

Worst-Case Monthly Profit

55.6% Margin

₹58,993

Based on 1,000 abuser users on the Yearly Plan. Total monthly COGS is ₹47,186.

Monthly Income Statement (1,000 Active Users)

FINANCIAL LINE ITEM	MONTHLY PLAN (REALISTIC MIX)	YEARLY PLAN (REALISTIC MIX)	MONTHLY PLAN (WORST-CASE)	YEARLY PLAN (WORST-CASE)
Gross Revenue	₹2,49,000.00	₹1,24,917.00	₹2,49,000.00	₹1,24,917.00
Google Play Comm. (15%)	-₹37,350.00	-₹18,738.00	-₹37,350.00	-₹18,738.00
Net Retained Revenue	₹211,650.00	₹106,179.00	₹211,650.00	₹106,179.00
Gemini AI Token Expense	-₹2,836.00	-₹2,836.00	-₹45,000.00	-₹45,000.00
Firestore Read/Write Expense	₹0.00 *(Free)*	₹0.00 *(Free)*	-₹423.75	-₹423.75
Cloud Functions Compute Expense	₹0.00 *(Free)*	₹0.00 *(Free)*	-₹1,762.25	-₹1,762.25
Net Operating Profit / Mo	+₹208,814.00	+₹103,343.00	+₹164,464.00	+₹58,993.00
Gross Profit Margin	98.7%	97.3%	77.7%	55.6%

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7-Day Trial Launch Simulation (25,000 Installs)

To evaluate marketing economics, we modeled a launch budget driving **25,000 installs in one week** with a **7-day free, unlimited trial**. Using standard mobile conversion rates, we analyzed the total trial infrastructure costs relative to the subscriber revenue generated at conversion.

Realistic Funnel (50% Active)

₹8,250 Total Cost

12,500 active trial users. Database costs fit within free tier.
Avg cost/user: ₹0.66.

High Funnel (100% Active)

₹16,684 Total Cost

25,000 active trial users logging data. Firestore writes billable. Avg cost/user: ₹0.67.

Investment Return (ROI) Analysis

Assuming a conservative industry standard of **1.5% to 2.0%** of the trial users converting to the ₹1,499 yearly subscription at the end of the trial period:

Metric	Low Conversion (1.5%)	High Conversion (2.0%)	Analysis & ROI Basis
Converted Subscribers	375 users	500 users	Derived from 25,000 trial installers.
Gross Revenue Generated	₹5,62,125	₹7,49,500	Paid upfront (100% yearly plan).
Net Retained Revenue (Play Store)	₹4,77,806	₹6,37,075	Net cash inflow to Zivofit.
Trial Infrastructure Costs	-₹16,684	-₹16,684	Worst-case high activation infrastructure cost.
Net Launch Profit	+₹461,122	+₹620,391	Profit after subtracting all server/AI costs.
ROI (Revenue-to-Server Cost)	28.6x	38.2x	Every ₹1 spent on AI/Firebase returns ₹28-₹38.

Investor Takeaway: High-Margin Scalability

Because backend operational costs are extremely low, Zivofit can offer low-friction, high-value free trials aggressively. Our marketing budget can focus entirely on customer acquisition (CAC) and conversion rate optimization (CRO), since backend infrastructure and AI costs represent a negligible **~2.6%** of net retained subscriber revenue.

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